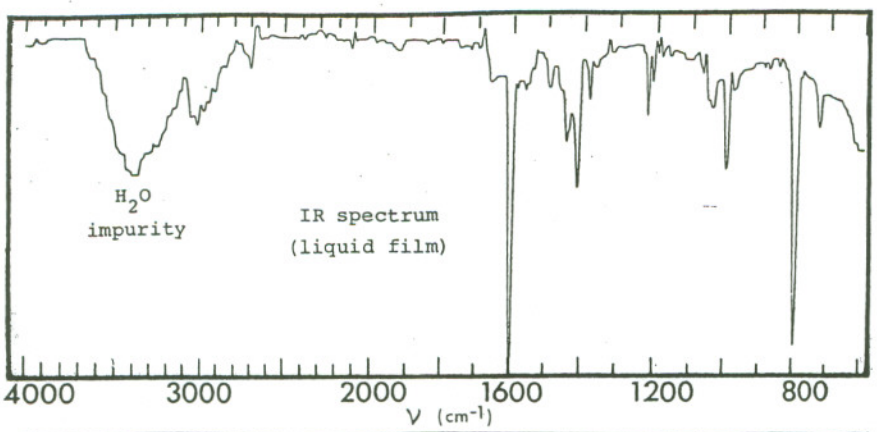
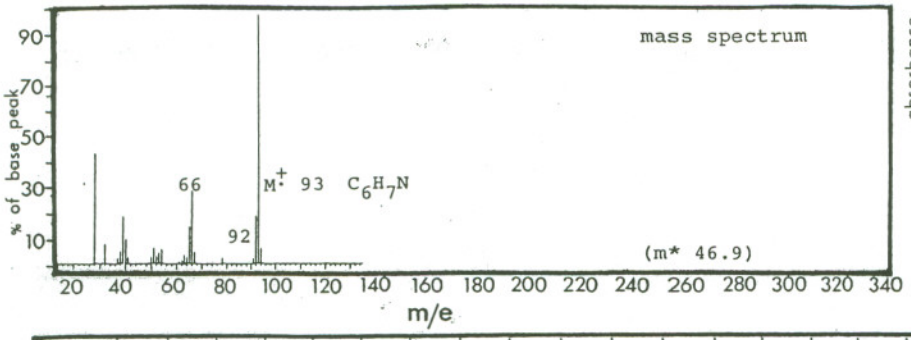
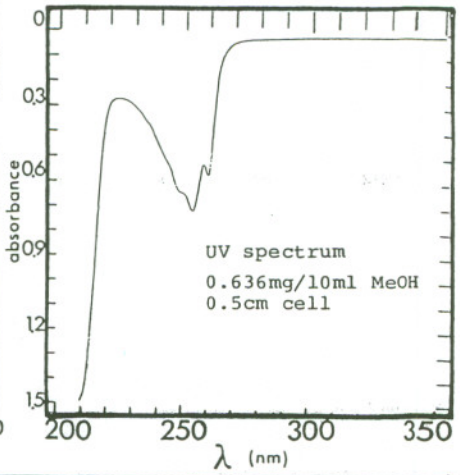
Organic Cume: Give the structures of the compounds that give the following spectra. Indicate how you made your decisions.

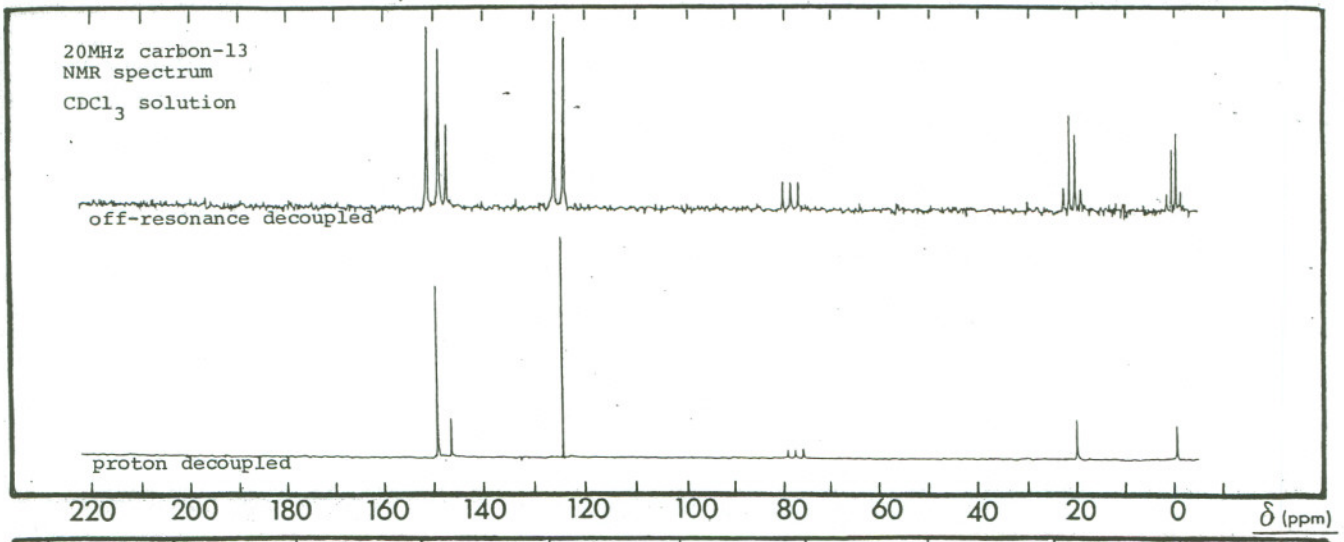
APRIL 22, 2005

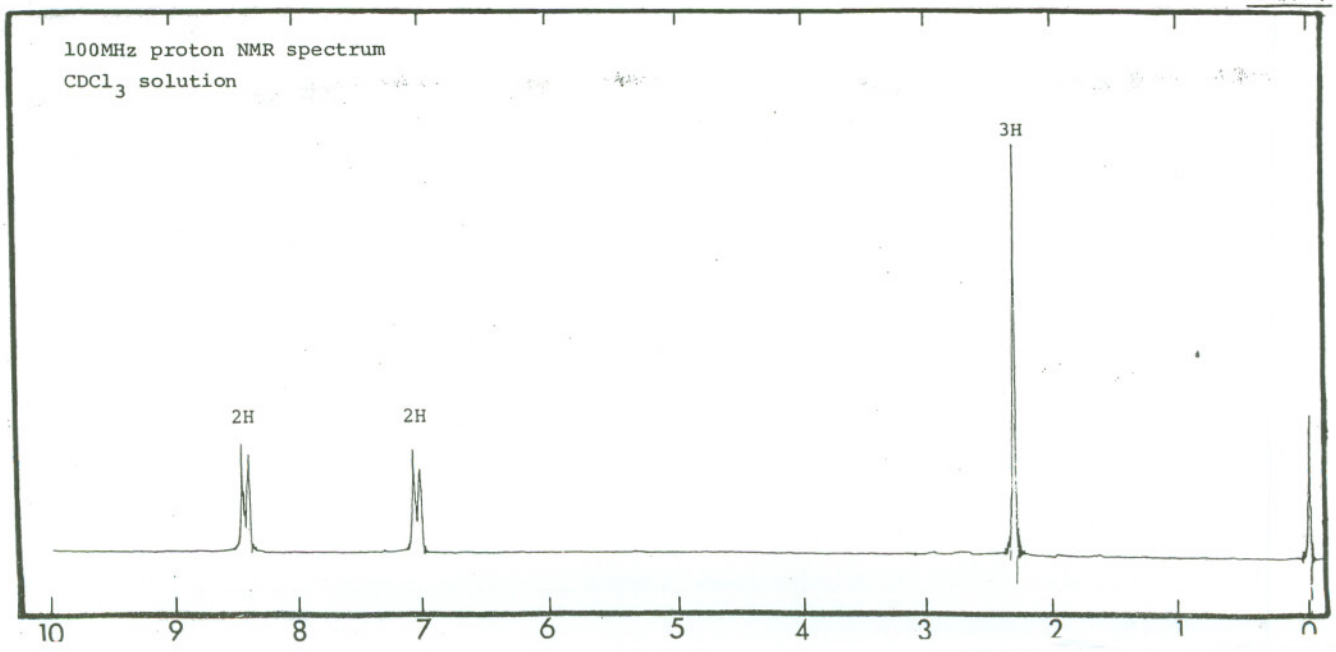
Problem 1



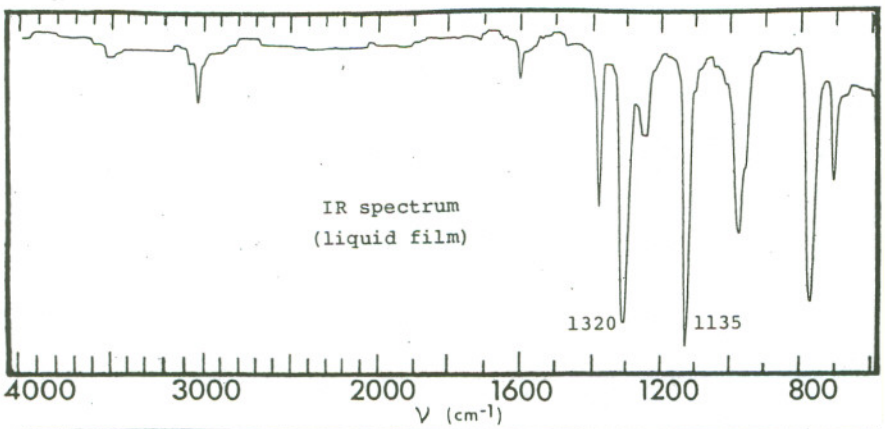


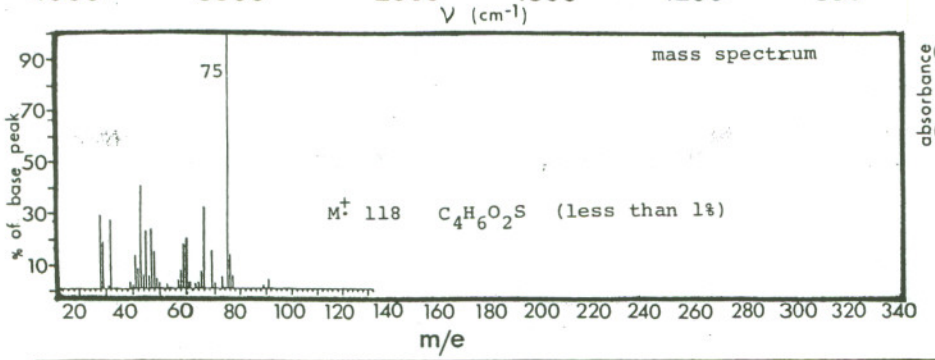


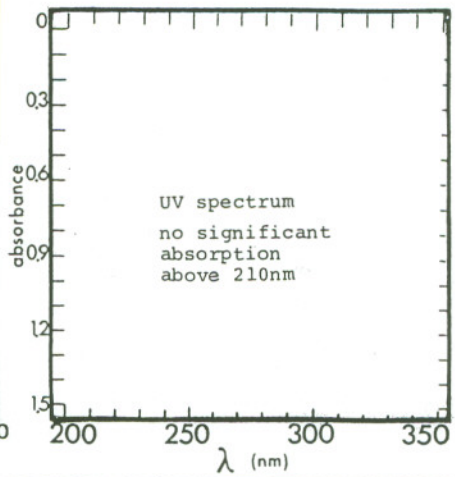


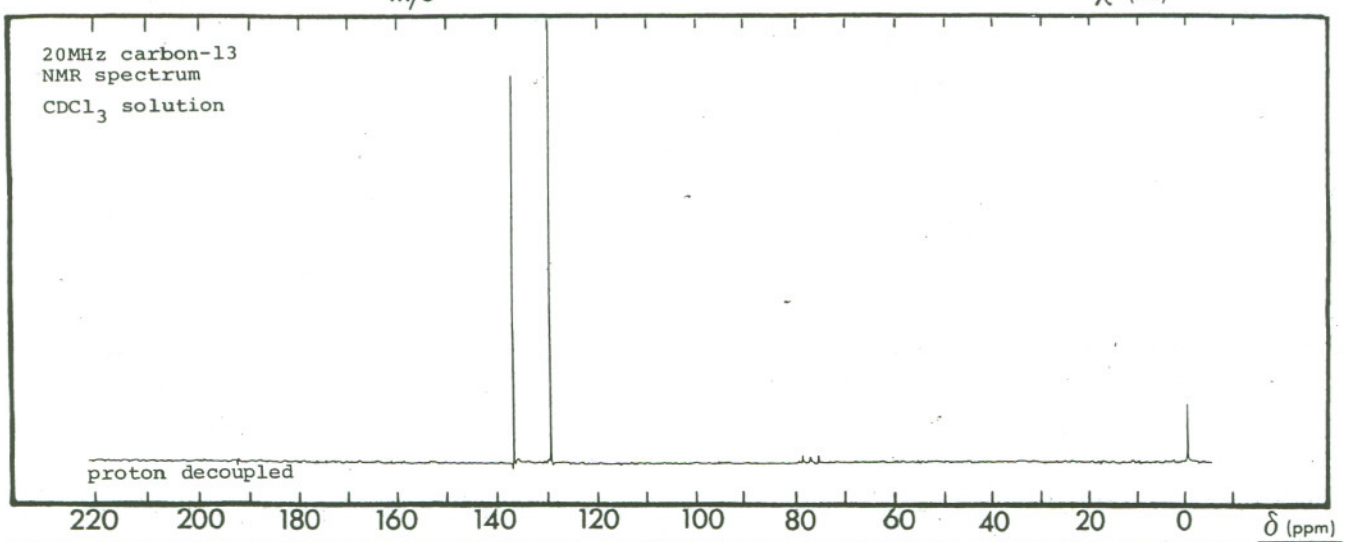


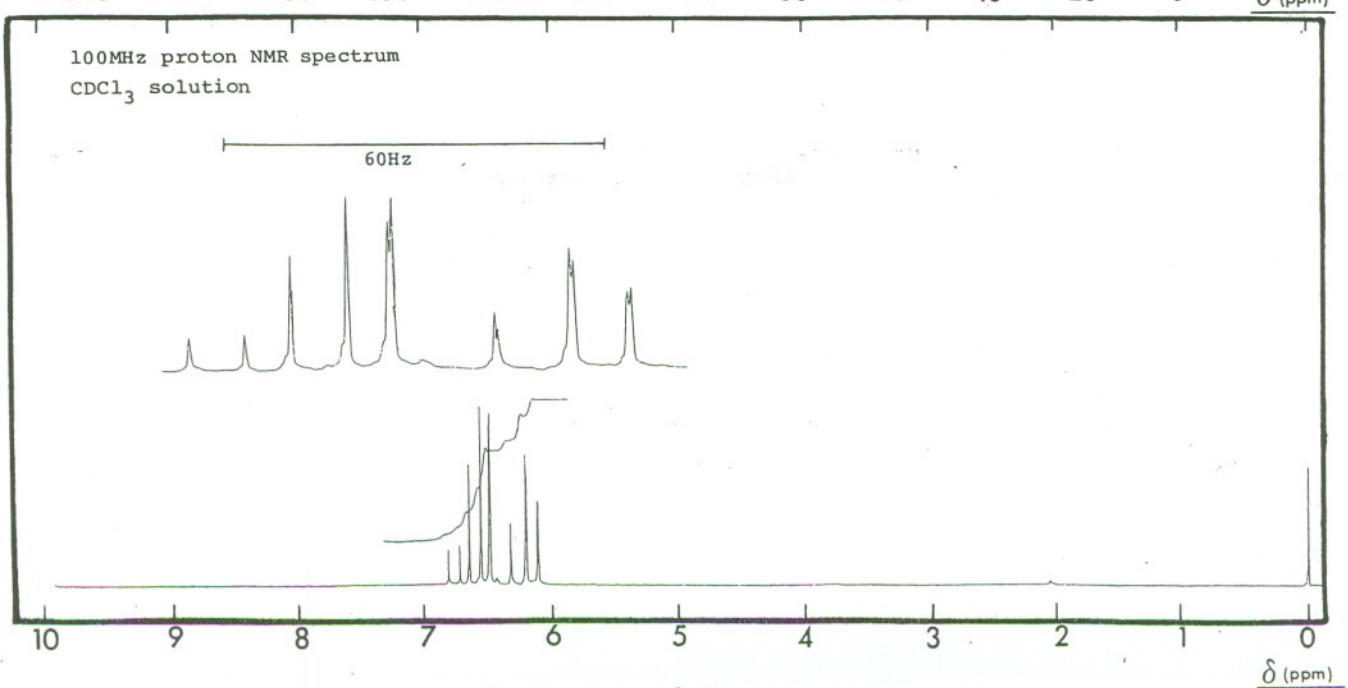
Problem 2


IR spectrum
(liquid film)
1320
1135
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

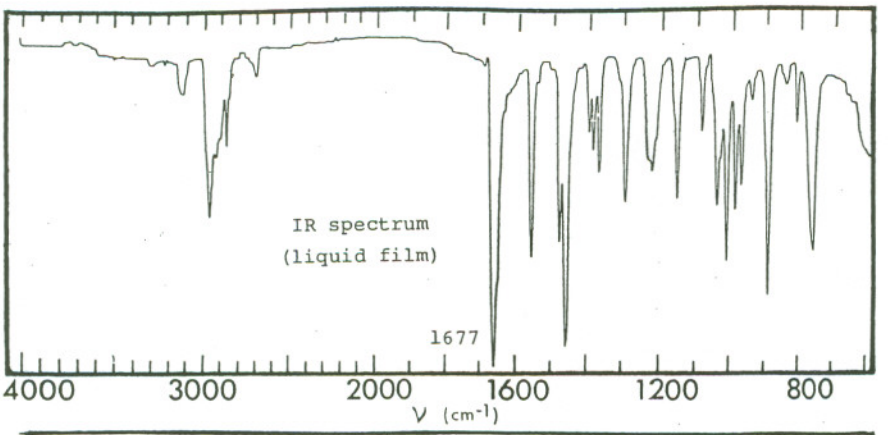

mass spectrum
75
% of base peak
M⁺ 118 C4H6O2S (less than 1%)
m/e

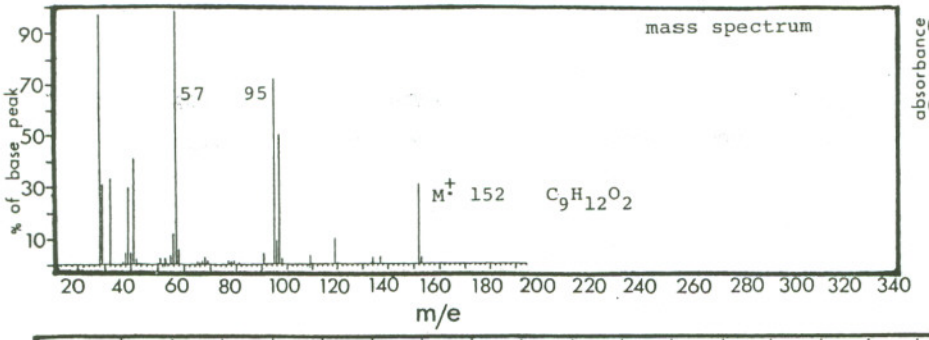

UV spectrum
no significant
absorption
above 210nm
absorbance
λ (nm)

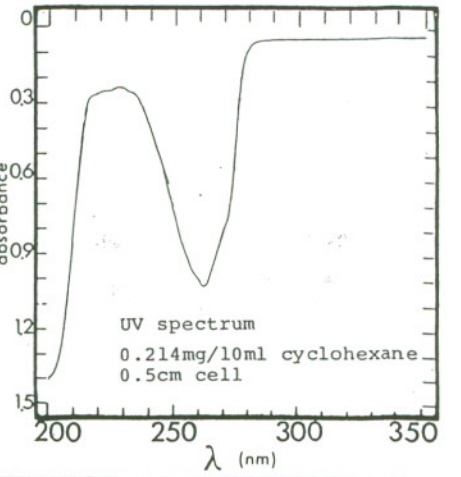

20MHz carbon-13
NMR spectrum
CDCl3 solution
proton decoupled
δ (ppm)

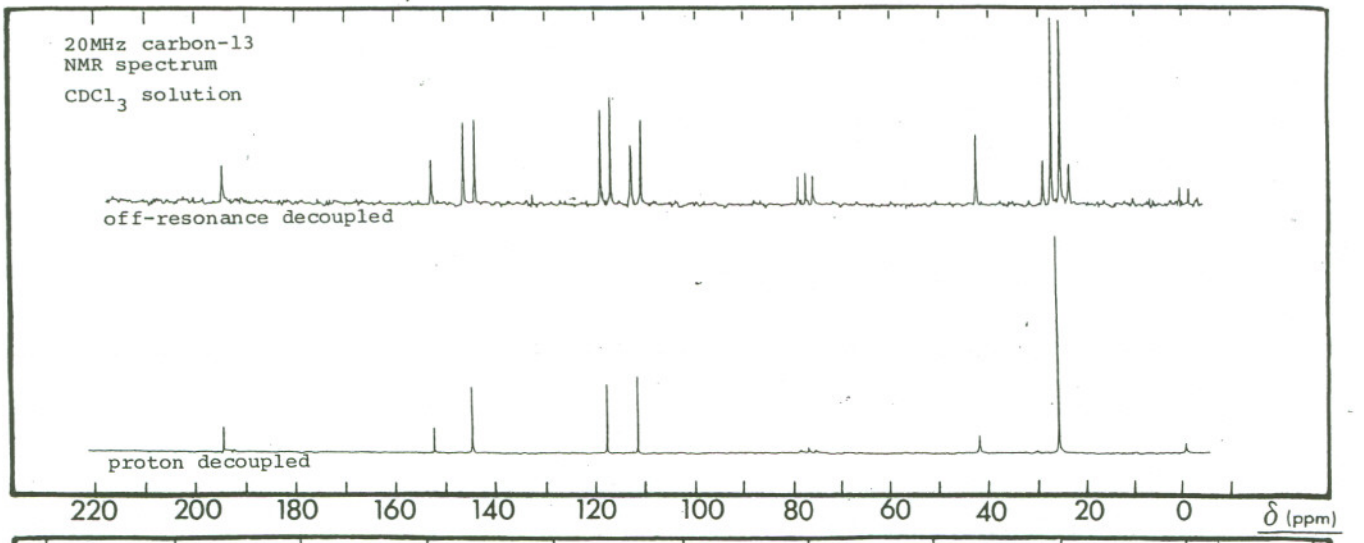

100MHz proton NMR spectrum
CDCl3 solution
60Hz
δ (ppm)

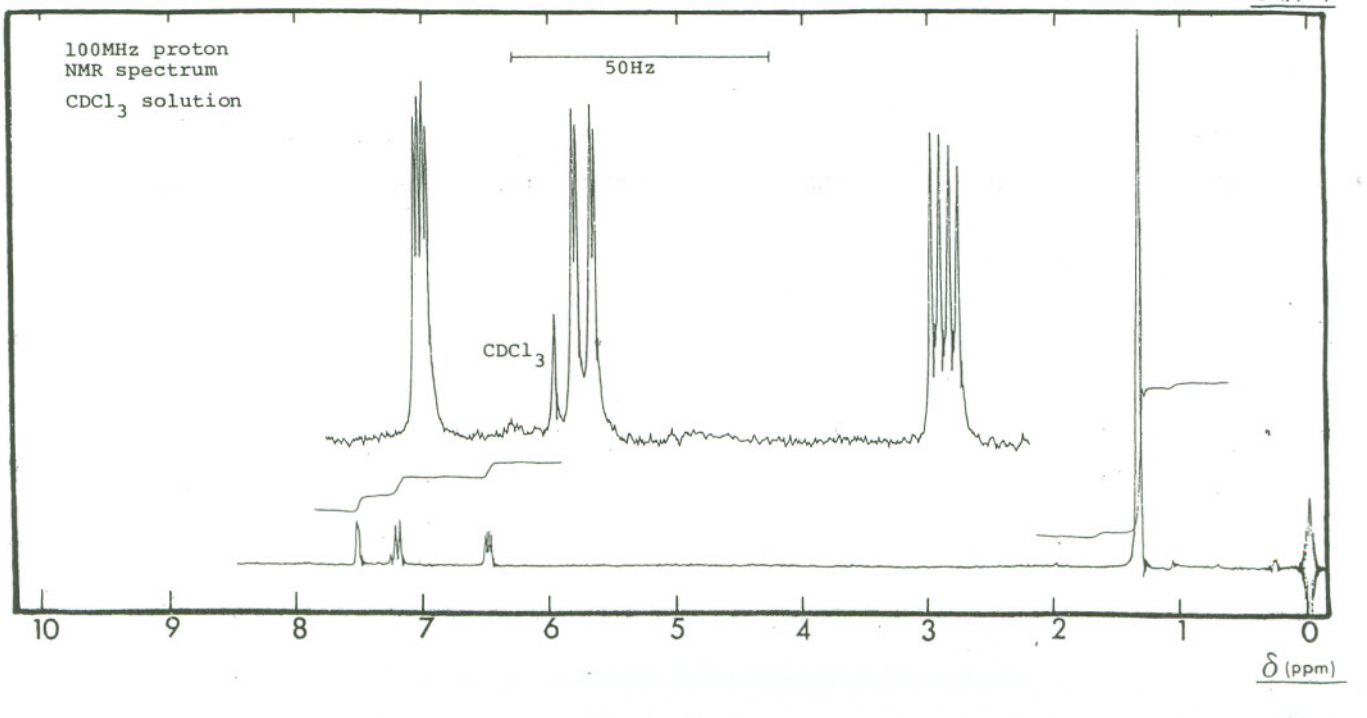
Problem 3


IR spectrum
(liquid film)
1677
4000
3000
2000
1600
1200
800
ν (cm⁻¹)


mass spectrum
% of base peak
57
95
M⁺ 152
C9H12O2
m/e


absorbance
UV spectrum
0.214mg/10ml cyclohexane
0.5cm cell
λ (nm)


20MHz carbon-13
NMR spectrum
CDCl3 solution
off-resonance decoupled
proton decoupled
δ (ppm)


100MHz proton
NMR spectrum
CDCl3 solution
50Hz
CDCl3
δ (ppm)